

Project Report: Local Office Communication System

1. Project Overview & Repository

The objective of this project is to deploy a secure local communication infrastructure, including an internal email server and a role-based file-sharing system.

- **GitHub Repository:** https://github.com/Leo-PHAV/EUTech_Leo-PHAV.git
- **Branch:** [task-local-office-Leo-PHAV](#)

2. Methodology & Technical Stack

- **Email Services:** **Postfix** (SMTP) for message routing and **Dovecot** (IMAP) for mailbox retrieval.
- **File Sharing:** **Samba** configured for secure network sharing across local clients.
- **Access Control:** Implementation of Role-Based Access Control (RBAC) using **Linux groups** (managers) and **Samba permissions**.

3. Configuration & Security

3.1 Email Policy

A strict message size limit was implemented to prevent server congestion.

- **Limit:** 10MB per message (configured via [message_size_limit](#) in Postfix).
- **Alias Management:** System aliases were configured in [/etc/aliases](#) to redirect administrative emails to [user1](#).

3.2 Role-Based Access Control (RBAC)

Two primary roles were simulated:

- **Managers Group:** Full access to the [\[Private\]](#) shared folder.
- **Staff Group:** Access restricted to the [\[Public\]](#) folder only.

```
leo-phav@leo-phav: ~
GNU nano 7.2 /etc/samba/smb.conf
# (you need to configure Samba to act as a domain controller too.)
# The path below should be writable by all users so that their
# profile directory may be created the first time they log on
;[profiles]
;   comment = Users profiles
;   path = /home/samba/profiles
;   guest ok = no
;   browseable = no
;   create mask = 0600
;   directory mask = 0700

[printers]
comment = All Printers
browseable = no
path = /var/tmp
printable = yes
guest ok = no
read only = yes
create mask = 0700

# Windows clients look for this share name as a source of downloadable
# printer drivers
[print$]
comment = Printer Drivers
path = /var/lib/samba/printers
browseable = yes
read only = yes
guest ok = no
# Uncomment to allow remote administration of Windows print drivers.
# You may need to replace 'lpadmin' with the name of the group your
# admin users are members of.
# Please note that you also need to set appropriate Unix permissions
# to the drivers directory for these users to have write rights in it
; write list = root, @lpadmin

[Public]
path = /srv/samba/public
browseable = yes
read only = no
guest ok = yes

[Private]
path = /srv/samba/private
valid users = @managers
browseable = yes
read only = no

^G Aide      ^O Écrire    ^W Chercher  ^K Couper    ^T Exécuter  ^C Emplacement M-U Annuler
^X Quitter   ^R Lire fich.^_ Remplacer  ^U Coller    ^J Justifier ^_ Aller ligne M-E Refaire
```

4. System Validation (Proofs)

4.1 Email Connectivity

Successful local communication between `user1@localhost` and `user2@localhost` was verified via Thunderbird.

Search...CTRL + K

Inbox - user1@localhost

Account Setup

Your full name

User Two

Email address

user2@localhost

Password

Welcome2026!

☒ Remember password

✓ The following settings were found by probing the given server:

Manual configuration

INCOMING SERVER

Protocol:

IMAP

Hostname:

.localhost

Port:

143

Connection security:

None

Authentication method:

Normal password

Username:

user2

OUTGOING SERVER

Hostname:

.localhost

Port:

25

Connection security:

None

Authentication method:

Normal password

Username:

user2

Advanced config

Re-test

Cancel

Done

Not sure what to select?

[Setup documentation](#) - [Support forum](#) - [Privacy policy](#)

Search...CTRL + K

Inbox - user2@localhost

Test

User One <user1@localhost>

user1@localhost

To user2@localhost

Test

Test mail local office com system

Reply

Forward

Archive

Junk

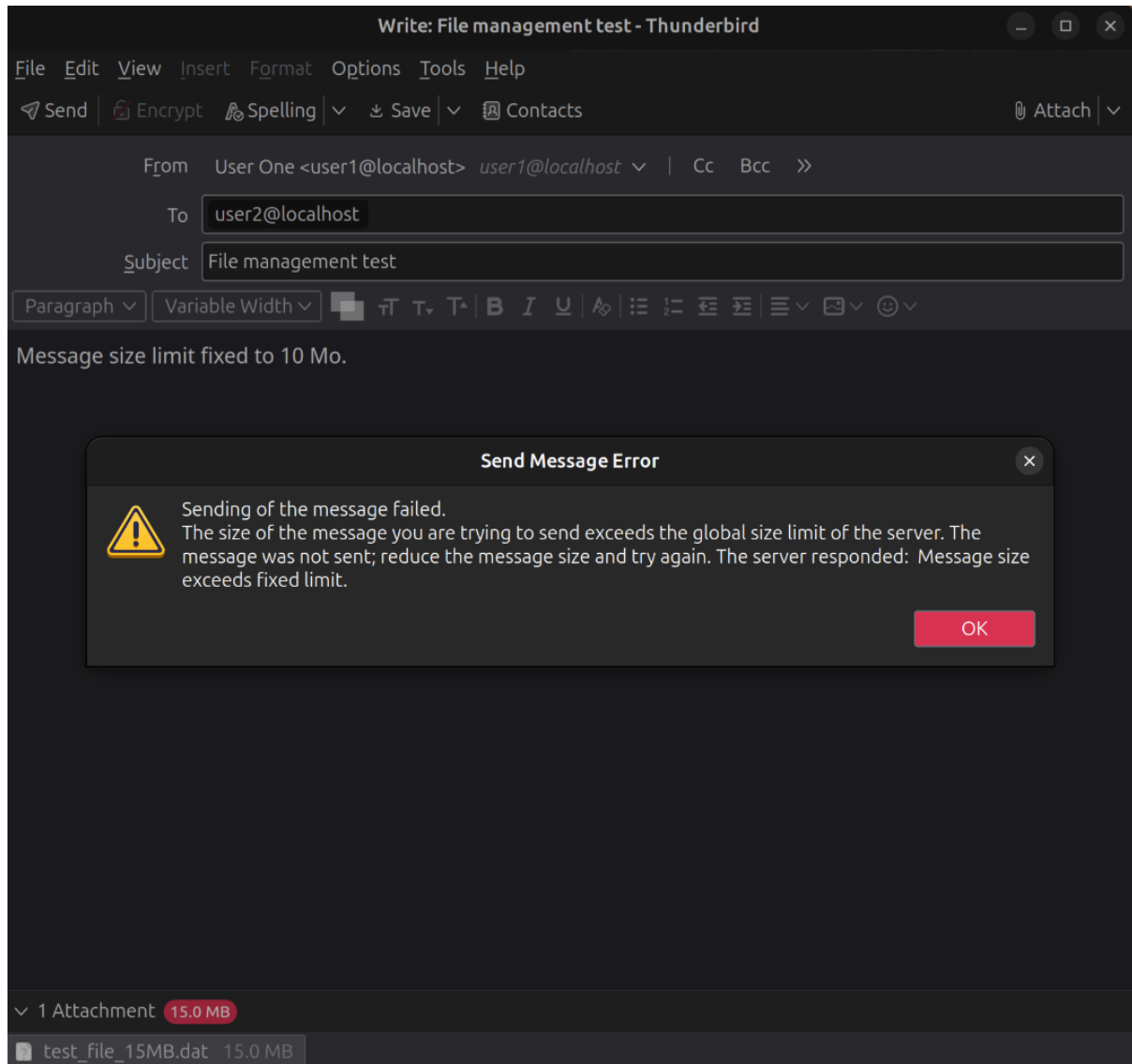
Delete

More

11:41

4.2 Security & Quota Testing

The attachment size limit was validated by attempting to send a 15MB file, which was correctly rejected by the server.



4.3 Samba Access Verification

The terminal was used to verify that **user2** is blocked from the management folder while **user1** maintains access.

```
leo-phav@leo-phav: ~  
leo-phav@leo-phav:~/Documents$ cd  
leo-phav@leo-phav:~$ smbclient //localhost/Private -U user1%Welcome2026! -c "ls"  
.  
D 0 Thu Feb 5 11:30:48 2026  
..  
D 0 Thu Feb 5 11:30:48 2026  
48899748 blocks of size 1024. 13280248 blocks available  
leo-phav@leo-phav:~$ smbclient //localhost/Private -U user2%Welcome2026! -c "ls"  
tree connect failed: NT_STATUS_ACCESS_DENIED  
leo-phav@leo-phav:~$
```

5. User Guides & Maintenance

5.1 Connection Manual

- **Email:** Use Thunderbird with server `localhost`, ports `143` (IMAP) and `25` (SMTP).
- **Files:** Connect via `smb://localhost/` using system credentials.

5.2 Service Management

The system services are managed via `systemctl` for reliability:

- `sudo systemctl restart postfix dovecot smbd`